

# Isla Stiven

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## EDUCATION

### Bachelor of Engineering in Mechanical Engineering (B.Eng)

University of Victoria | Thermo-fluids and Aerodynamics Specialization | GPA: 3.8/4.0

Expected Graduation: December 2026

Victoria, BC

## SKILLS

**Design Skills** CAD | Design for Manufacturing | FEA | CFD | Rapid Prototyping | PCB Design | GD&T | Instrumentation | Engineering Drawing | Shop Tools | Embedded Systems | Fluid and Pneumatic Systems

**Technologies** SolidWorks | Siemens NX | ANSYS Granta EduPack | KiCAD | LabVIEW | MATLAB | Arduino | Python | C++

## EXPERIENCE

### ARTMS Inc.

Mechanical/Mechatronics Engineering Intern

January 2025 - August 2025

Burnaby, BC

- Owned the **design, build, and test** of a device used to measure internal component temperatures during **cyclotron irradiation**, engineered to meet strict requirements such as cyclotron interface compatibility, a fully sealed enclosure, and zero-interference with the irradiation process to improve understanding of **thermal failure modes**.
- Developed and tested a **printed circuit board (PCB)** for ARTMS's **nickel-electroplating unit**, enabling efficient in-house manufacturing of solid targets for **radioisotope production**.
- Employed **geometric dimensioning and tolerancing (GD&T)** on engineering drawings to define part tolerances, ensuring clear communication with **machinists**.
- Designed, conducted, and documented **experimental tests** to validate improvements in isotope production, increasing **cooling flow 30%** through Swagelok-fitted water systems, **autonomizing target production** via refined target-polishing methods, and mitigating system wear through **radiation-hard material selection**.
- Assembled, integrated, and tested** multiple major subsystems of the ARTMS **GEN2 QIS**, including **sensor/actuator** integration and fluid and **pneumatic systems**, deployed on cyclotrons to increase yield and improve efficiency of radioisotope production at **international sites**.

### University of Victoria SEGAL Lab

Software Engineering Intern

September 2023 - December 2023

Victoria, BC

- Architected and built a cross-platform (iOS/Android) educational app prototype in **Flutter/Dart** with a 4-student start-up team, successfully delivering a functional MVP for teachers in remote communities, earning a **CBC interview** and an invite to the AISES National Conference.

## PROJECTS AND INVOLVEMENT

### Air Quality/AQHI+ Monitor

Personal Project

August 2026 - Present

Victoria, BC

- Engineered an **air quality monitor** that fuses particulate, CO<sub>2</sub>, and VOC sensor data to compute a live Air Quality Health Index (**AQHI**) for wildfire smoke events, pairing **custom C++ firmware** with an **airflow-optimized 3D-printed enclosure** designed for accurate sampling.

### Wingfoil Handle Force Measurement System

Tagmata Materials

January 2026 - Present

Victoria, BC

- Designed, simulated, built, and tested a functional **force measurement system** for wing foil athletes using **dual inline load cells** mounted to a **carbon fiber** auxiliary boom, enabling independent front and rear hand force capture during live water sessions to generate data for improving wing **aerodynamics and efficiency**.
- Developed a **sealed, IP67-rated electronics package** built around an **ESP32-S3** microcontroller, dual load-cell **ADC amplifiers**, a **9-DOF IMU**, and a **LiPo power system**, logging **time-synchronized force and orientation data** at **50 Hz** with **±5% FS accuracy** for reliable acquisition in saltwater environments.

### FSAE Suspension A-Arm Nodes

UVic Formula Racing Club

January 2026 - April 2026

Victoria, BC

- Designed four aluminum suspension A-arm nodes in SolidWorks using **FEA and topology optimization** to achieve 64–70% **mass reduction**, then applied **design-for-manufacturing principles** to produce **CNC-machinable** nodes bonded to carbon fiber tubes, with adhesive joints sized to carry simulated suspension loads.

### The Victoria Hand Project

Volunteer Prosthetic Hand Technician

January 2023 - January 2024

Victoria, BC

- Fabricated and assembled functional upper-limb prosthetics from 3D-printed and laser-cut components, developing clearer assembly **documentation** to standardize the build process.

## HOBBIES

Running, Hiking, Alpine and Nordic Skiing, Piano, Cycling, Photography